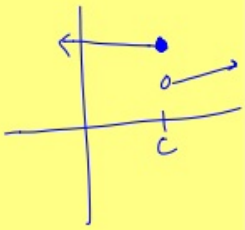
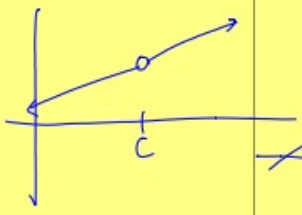
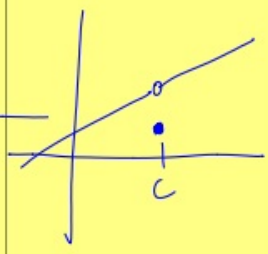
LIM-2.A.2 A function f is continuous at $x = c$ provided that $f(c)$ exists, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$.
LIM-2.B.1 A function is continuous on an interval if the function is continuous at each point in the interval.
LIM-2.B.2 Polynomial, rational, power, exponential, logarithmic, and trigonometric functions are continuous on all points in their domains.
LIM-2.C.1 If the limit of a function exists at a discontinuity in its graph, then it is possible to remove the discontinuity by defining or redefining the value of the function at that point, so it equals the value of the limit of the function as x approaches that point.
LIM-2.C.2 In order for a piecewise-defined function to be continuous at a boundary to the partition of its domain, the value of the expression defining the function on one side of the boundary must equal the value of the expression defining the other side of the boundary, as well as the value of the function at the boundary.

1-11 to 1-13 Continuity

Definition of Continuity at a Point	A function f is **continuous at $x = c$** when three conditions are met. 1. $f(c)$ is defined 2. $\lim_{x \to c} f(x)$ exists 3. $\lim_{x \to c} f(x) = f(c)$ **Note**: For Condition 2, when using a piecewise function, we must find the limit from **BOTH** sides to verify that the limit exists.

Example 1: Sketch a graph with the following conditions.

Condition 1 is true but **Condition 2** does NOT hold.	**Condition 2** is true but **Condition 1** does NOT hold.	**Conditions 1 and 2** are true but **Condition** 3 does NOT hold.
		

A function is <u>continuous from the left</u> at x = c	A function is <u>continuous from the right</u> at x = c
$\lim_{x \to c^-} f(x) = f(c)$	$\lim_{x \to c^+} f(x) = f(c)$

Example 2: Show that the following function is continuous at $x = 1$.

$$f(x) = \begin{cases} 2x^2 - 3, & x < 1 \\ \ln(x) - e^{x-1}, & x \geq 1 \end{cases}$$

$f(1) = \ln 1 - e^{1-1} = 0 - e^0 = 0 - 1 = -1$

$\lim_{x \to 1} f(x) = -1$

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (2x^2 - 3) = 2 - 3 = -1$

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (\ln x - e^{x-1}) = \ln 1 - e^{1-1} = -1$

$f(x)$ is cont. at $x = 1$ b/c $\lim_{x \to 1} f(x) = f(1)$

Example 3: Is the following function continuous at $x = 3$? Give a reason for your answer.

$$g(x) = \begin{cases} x^2 - 5, & x < 3 \\ 9 - 2x, & x \geq 3 \end{cases}$$

$\overline{}$
3

$g(3) = 9 - 2(3) = 9 - 6 = 3$

$\lim_{x \to 3} g(x) = \text{DNE}$

$\lim_{x \to 3^-} g(x) = \lim_{x \to 3^-} (x^2 - 5) = 9 - 5 = 4$

$\lim_{x \to 3^+} g(x) = \lim_{x \to 3^+} (9 - 2x) = 9 - 6 = 3$

$g(x)$ is discontinuous at $x = 3$ b/c $\lim_{x \to 3} g(x)$ DNE

Example 4: Find a value for c such that the function $k(x)$ is continuous for all real numbers.

$$k(x) = \begin{cases} cx^2 - 2x, & x < 3 \\ 5 + c, & x \geq 3 \end{cases}$$

$K(x)$ is cont. at $x=3 \Rightarrow \lim_{x\to 3} k(x) = k(3)$

$$\lim_{x\to 3^-} k(x) = \lim_{x\to 3^-}(cx^2-2x) = 9c-6$$

$$k(3) = \lim_{x\to 3^+} k(x) = \lim_{x\to 3^+}(5+c) = 5+c$$

$$\lim_{x\to 3^-} k(x) = \lim_{x\to 3^+} k(x)$$

$$9c-6 = 5+c$$

$$8c = 11$$

$$\boxed{c = \frac{11}{8}}$$